

# **PROJECT PROPOSAL**

## **Developing a Standard Evaluation Framework for Event-Centric Knowledge Graphs**

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# Outline

- **Introduction to Event Knowledge Graphs**
- **Importance of EKG Evaluation**
- **Current Challenges**
- **Research Question**
- **Proposed Evaluation Framework**
- **Methodology**
- **Expected Outcomes**
- **Potential Challenges**

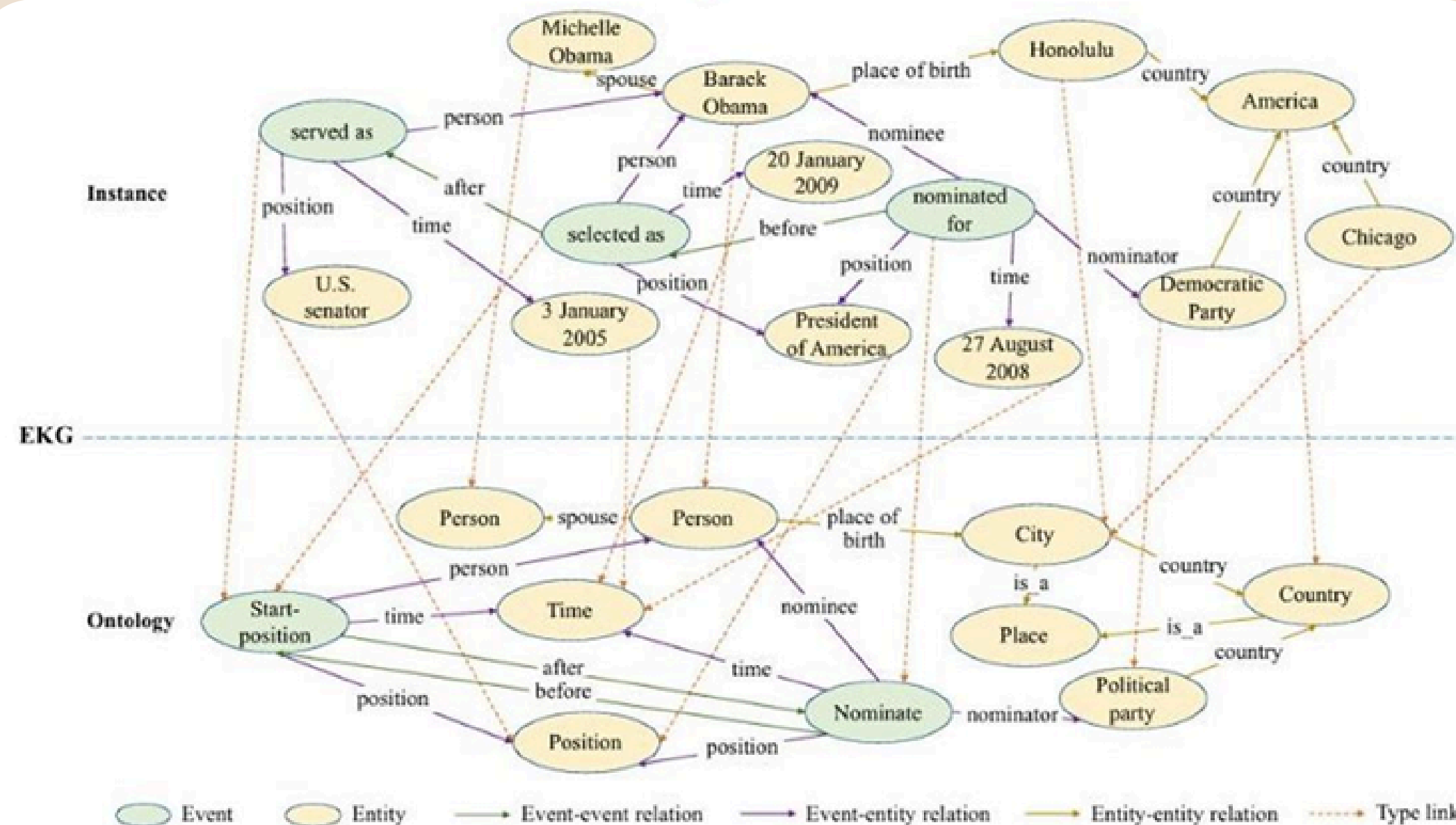
# EVENT KNOWLEDGE GRAPHS

## What are EKGs?

- Advanced data structures centered around events.
- Emphasize temporal, causal, and contextual connections.

## Significance:

- Crucial for event prediction and causal analysis.
- Applicable in news analytics, historical data, etc.



The EKG in the image illustrates the career progression of Barack Obama, capturing key events, roles, and relationships, with contextual details like birthplace and party affiliation.

# TRADITIONAL KNOWLEDGE GRAPHS

## Entity-Centric:

Focus on static entities like people, places, and organizations.

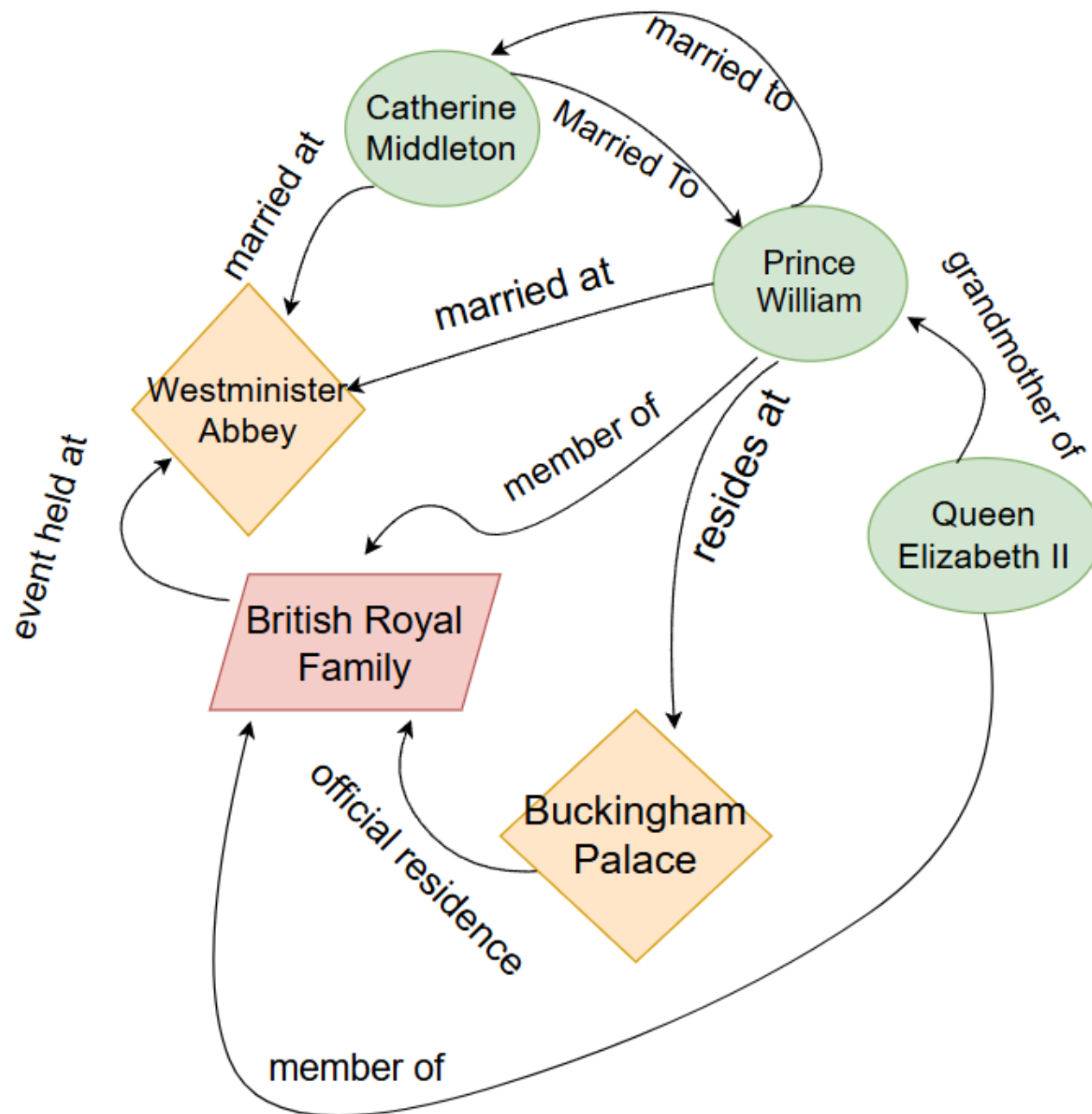
Represent relationships

## Static Relationships:

Lack inherent temporal or causal dimensions.

## Applications:

Semantic search, question answering, information retrieval.



# Event-Centric Knowledge Graphs

## Event-Centric:

Focus on events as primary nodes.

Emphasize the dynamic relationships between events and entities.

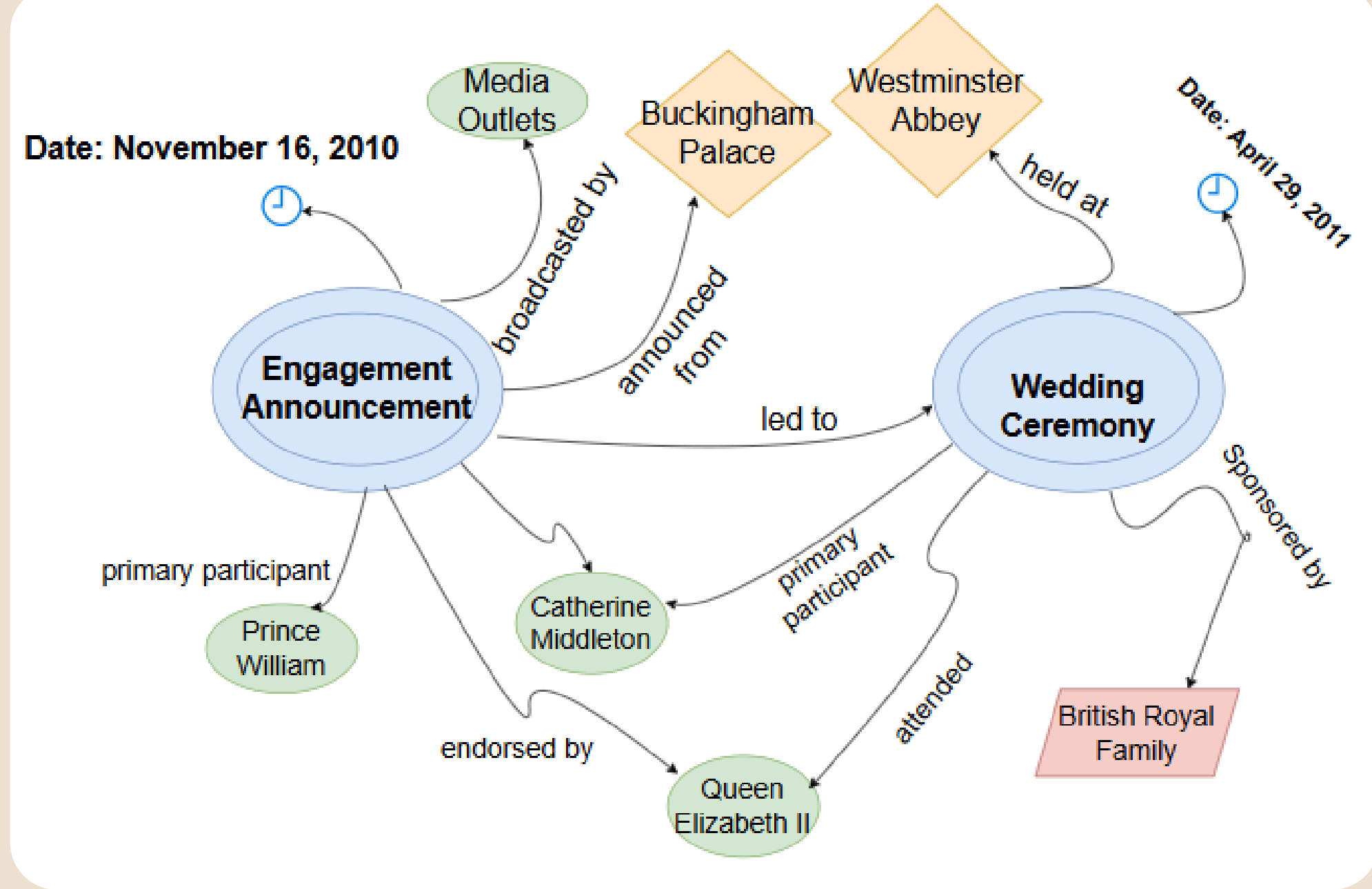
## Dynamic Relationships:

Capture temporal (when), causal (why), and spatial (where) aspects.

(also Contextual , and Spatio-temporal )

## Applications:

Event prediction, causal analysis, understanding event sequences.



# WHY EKG EVALUATION IS IMPORTANT

- Rigorous evaluation of EKGs is essential to ensure their effectiveness and reliability in real-world applications.
- Without thorough assessment, EKGs may provide misleading insights, leading to faulty decision-making in critical domains

**A comprehensive evaluation verifies how accurately EKGs model events, their many dynamics, and interdependencies—vital factors for tasks like event prediction and causal analysis.**

**Comprehensive evaluation is crucial to ensure that EKGs effectively handle complex event dynamics, such as causal relationships and temporal sequences, which are essential for accurate event prediction and strategic decision-making.**

- **Fraud Detection:** EKGs can model event sequences in transaction data [1]
- **Disease Outbreak Monitoring:** EKGs can track patterns and relationships between medical events aiding in outbreak prediction. [2]
- **Optimising Operations:** EKGs help model dependencies between supply chain events[3]

# EFFECTS OF EVALUATION, A TALE OF TWO EVALUATIONS

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Evaluation from "Building Event-Centric Knowledge Graphs from News"[4]

**Objective:** Focused on precision and recall to assess event representation accuracy.

## **Methodology:**

- Evaluated 1,043 triples sampled from 100 Wikinews articles.
- Relied on human judgment due to the lack of a gold standard.

## **Findings:**

- Overall triple accuracy: 55.1%.
- Accuracy improved to 69.7% when excluding coreferring events with incorrect mentions.

## **Limitations and Impact**

- **Narrow Evaluation Scope:**
  - Precision and recall overlook overall completeness and coverage.
- **Implications:**
  - Potential inaccuracies and reduced performance.
  - Systems relying on this ECKG risk decreased reliability and effectiveness.

Evaluation from "EventKG: A Multilingual Event-Centric Temporal Knowledge Graph": [5]

- **EventKG Evaluation:** Used a comprehensive set of metrics to assess coverage, completeness, and data quality
- **Key Metrics:** Total events and relations, temporal interlinking quality, event fusion consistency, textual descriptions
- **Outcome:** Highlighted EventKG's strengths in providing extensive, multilingual, event-centric temporal knowledge.

**Impact:** Ensures high reliability, completeness, and superior system performance, effectively meeting diverse user needs.

- **Limitations:**
- **Neglected Granularity and Hierarchical Event Representation:** Did not evaluate the ability to represent events at different levels of detail or establish hierarchical relationships.
- **Overlooked Multimodal Data Integration:** Omitted assessment of integrating data from multiple modalities (e.g., text, images, or audio), limiting insights into rich event representations.

# CURRENT EKG EVALUATIONS

- EKG evaluations often rely on precision, recall, and F1 scores, focusing narrowly on event relationships while overlooking broader system features like semantic richness and multilingual adaptability.
- Frameworks such as Discovering Event Evolution Graphs from News Corpora depend heavily on **manually annotated** relationships and predefined event sets, limiting their applicability.
- These evaluations make assumptions that restrict their ability to capture evolving, complex, and context-rich event systems effectively.
- Fragmented evaluation approaches fail to capture critical EKG features such as causality, event evolution, and interconnectivity.
- The absence of comprehensive frameworks limits understanding of EKG's.
- This gap undermines efforts to enhance EKGs' effectiveness across various domains.

**EKG evaluation methodologies are often disjointed, lacking an integrated framework to holistically assess their multifaceted nature.**

**While certain models, such as EventKG, explore completeness and coverage of temporal information, others, such as OEKG and ASER, either overlook key evaluation metrics or focus on isolated aspects without tying them into a cohesive analysis.**

# PROBLEM

Despite their growing significance, current evaluation methods are often fragmented, addressing only a subset of the features that define an EKG. For instance, existing methods may focus on the accuracy of event representation or temporal relationships but fail to provide a comprehensive assessment that considers all the critical dimensions of an EKG, such as causal links, contextual information, and the ability to handle complex queries involving multiple events.

## **In Short .....**

**There is a lack of a standardized and comprehensive evaluation framework to fully capture the features of EKGs.**

# RESEARCH OBJECTIVE

## Primary Aim

**Develop a comprehensive evaluation framework for EKGs**

## Specific Objectives

- **Integrate existing evaluation methods (Precision, recall, temporal).**
- **Introduce new metrics for reasoning and inference capabilities, Event Granularity, Contextual Completeness... and more**
- **Enhance accuracy and relevance of EKG evaluations.**

# PRIOR WORK ON KG EVALUATION FRAMEWORKS

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- **KGrEaT Framework:[6]**
- **Evaluation Method:** Focuses on task-based metrics (classification, clustering, recommendation) instead of intrinsic metrics like accuracy and completeness.
- **Strengths:** Modular design allows for easy extension with new tasks and datasets, offering flexibility and practical insights.
- **Limitations for EKGs:**
  - Ineffective for dynamic, temporal, and causal relationships in EKG's.
  - Requires significant adaptations to handle EKG-specific complexities.

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- **Chen et al.'s Framework:[7]**
- **Methodology:** Utilises scalable approaches like schema validation and sampling to meet practical application needs.
- **Strengths:** Effective for evaluating static knowledge graphs.
- **Limitations for EKGs:**
- **Lacks metrics for temporal reasoning, event sequence accuracy, and causal relationships.**
  - Completeness evaluation does not address the evolving nature of EKGs.
  - Overlooks coherence and reasoning capabilities, critical for dynamic event-based systems.

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# PROPOSED APPROACH

## Framework Design

- **Identify essential evaluation criteria**
- **Define core evaluation dimensions - features for EKG's**
- **Develop a conceptual model:** Establish relationships among evaluation criteria to guide metric development.

## Development of Metrics

- **Integrate existing metrics:** Use Precision, Recall, Temporal Consistency, and other metrics
- **Introduce novel metrics: Introduce metrics for measuring :**Reasoning and inference capabilities, Event Granularity, Contextual Completeness... and more
- **Focus on utility:** Ensure metrics evaluate practical applications like event prediction and risk assessment.

## Implementation

- **Develop algorithms and models for (EKG) analysis**
- **Create evaluation workflows:** Define step-by-step processes for applying criteria, running metrics, and interpreting results.
- **Build a framework implementation tool:** Include a user-friendly interface to generate evaluation reports for EKGs.

# METHODOLOGY

## Systematic Analysis

Analyse current evaluation methods for EKGs, focusing on strengths and gaps.

## Testing and Iteration:

**Prototype testing with benchmark EKGs.**  
**Iterative refinement based on feedback.**

## Create Evaluation Workflows:

Design step-by-step workflows detailing how to apply the evaluation criteria, run metrics, and interpret results for consistent assessment processes.

## Prototype Testing with Benchmark EKGs:

- Select Test Datasets: Use well-known EKGs like EventKG or ASER for testing.
- Run Evaluations:
- Apply the framework to these EKGs to see if it works as expected.
- Identify Issues: Look for any problems or inaccuracies in the results.
- Iterative Refinement:

## Perform Comparative Analysis:

- Compare with Other Methods: Evaluate the same EKGs using traditional methods.
- Analyze Differences: See how the results from the new framework differ from the old methods.
- Highlight Improvements: Point out areas where the new framework provides better or more detailed insights.

## Framework Development

### Identify Essential Evaluation Criteria

Review the unique characteristics of EKGs to pinpoint essential evaluation criteria

### Define Evaluation Dimensions:

Establish core dimensions

Outline specific aspects to measure within each dimension,

### Integrate Existing Metrics:

Incorporate well-established metrics like Precision, Recall, Temporal Consistency into the framework to assess various EKG features holistically.

### Incorporate metrics for other features often neglected by EKG evaluations:

- Expand the framework to include additional evaluation criteria that address underrepresented aspects of EKG performance.
- Ensure the metrics comprehensively capture the multidimensional nature of EKG features and their applications.

## Comprehensiveness of Evaluation

How well does the framework assess the full range of EKG features, including temporal, causal, spatial, and contextual dimensions?

Does it evaluate both event-entity and event-event relationships effectively?

## EVALUATION QUESTIONS

## Causal Relationships

How effectively does the framework assess the accuracy and depth of causal connections between events?

Can it evaluate indirect or complex causality?

## Contextual Richness

How robust is the evaluation of contextual integration, such as location, participants, and event-specific attributes?

Does it measure the semantic richness and completeness of contextual information?

## Evaluation Metrics Validation

How does the framework ensure the validity and reliability of its metrics?

Are the metrics meaningful, measurable, and aligned with real-world outcomes?

## **Comparison Across Models**

- How effectively does the framework compare different EKG models?
- Does it highlight strengths and limitations across models using standardised benchmarks?

# EVALUATION

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## Objectives

- **Validate the framework's ability to assess key EKG dimensions, including temporal accuracy, causal inference, and contextual richness.**
- **Demonstrate the robustness of new metrics such for features such as event granularity, contextual completeness and more**
- **Benchmark the framework against existing evaluation methods, showcasing its advantages.**

## Outcome

- **A comprehensive understanding of the framework's performance, including its strengths, limitations, and areas for improvement.**
- **Benchmarks for **accuracy**, **efficiency**, and **usability** compared to existing methods.**

# EVALUATION

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## Methodology

- **Phase 1: Prototype Testing**
  - Test the framework with benchmark datasets (EventKG, AugGKG, ASER).
  - Assess performance across dimensions such as temporal consistency, contextual richness, and task-specific adaptability and others.
- **Phase 2: Iterative Refinement**
  - Refine metrics and algorithms based on initial testing results.
  - Ensure flexibility to accommodate diverse EKG types and applications.
- **Phase 3: Comparative Analysis**
  - Compare the framework with traditional evaluation methods on multiple EKG models.
  - Highlight advantages in capturing EKG complexities like causal relationships and multimodal integration.

# POTENTIAL CHALLENGES

## ● Development of Novel Metrics:

- Defining meaningful and applicable metrics.
- Defining TSPS and TAR requires aligning theoretical and practical aspects for universal applicability.

## ● Data Availability and Quality

- Limited access to high-quality datasets like EventKG can restrict robust testing and validation.
- Access to high-quality benchmark datasets.

## ● Comparative Analysis Difficulties

- Aligning evaluation criteria with existing methods poses challenges in ensuring fairness and consistency.
- Aligning evaluation criteria across methods.

## ● Validation Across Domains:

- Balancing domain-specific requirements while maintaining framework standardization is complex.
- Balancing diverse domain requirements.

# EXPECTED OUTCOMES

## Comprehensive Framework

- Standardized evaluation for EKGs.

## Novel Metrics

- Task-Specific Performance Score (TSPS).
- Task Adaptability Rate (TAR).

## Advancement in the Field

- Improved understanding of EKG strengths and limitations.
- Enhanced comparability of EKG models

# CONCLUSION

## Summary

- **Identified Gap:** Lack of a standardized and comprehensive evaluation framework for Event-Centric Knowledge Graphs (EKGs).
- **Research Objective:** Develop a comprehensive evaluation framework that integrates existing methods and introduces novel metrics tailored to EKGs.

## Expected Contributions

- Comprehensive Framework: **Provide a standardized method to evaluate EKGs holistically.**
- Novel Metrics: Introduce metrics like to assess contextual ,granularity-focused nuances, and more....
- Advancement in the Field: **Enhance understanding and comparability of different EKG models**, leading to more reliable applications.

## Next Steps

- Methodology Implementation: Conduct systematic analysis, develop the framework, and test with benchmark EKGs.
- Validation and Refinement: Iterate based on testing outcomes and expert feedback to refine the framework and metrics.
- Address Potential Challenges: Tackle issues like data availability and ensure adaptability across domains.

## Anticipated Impact

- Improved EKG Evaluation: Enhance the accuracy and relevance of evaluations, benefiting various domains.
- Support for Decision-Making: Provide a reliable foundation for systems that depend on EKGs for critical insights.

# REFERENCES

- [1] Dynamic Knowledge Graph Based Multi-Event Forecasting, S Deng et al., 2020
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- [7] Zhu, X., Qin, B., Zhu, X., Liu, M., & Qian, L. (Eds.). (2019). Knowledge Graph and Semantic Computing: Knowledge Computing and Language Understanding. Communications in Computer and Information Science.